

Discrete Fourier Transformation and Quantum Fourier Transformation

Introduction:

- ◆ **What is Fourier Transformation?**

A Fourier Transformation is a mathematical method that transforms a signal from its original domain (like time or space) into the frequency domain.

- ◆ **Why we Use It?**

- To break down complex signals or images into simpler wave components.
- Useful in **signal processing, image compression, communication systems, watermarking** , etc.

◆ **This Presentation Covers:**

- **DFT** – Discrete Fourier Transform (Classical)
- **Inverse DFT** – Getting back the original signal
- **QFT** – Quantum Fourier Transform (Quantum version)
- **Inverse QFT** – Reverses QFT to retrieve input state

Discrete Fourier Transformation(DFT)

- Discrete Fourier Transformation converts the time domain signal into a frequency domain signal.
- Formula:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) * e^{-i2\pi(xu/M + yv/N)}$$

Euler's formula:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

when $\theta = \pi$

$$\begin{aligned} e^{i\pi} &= \cos \pi + i \sin \pi \\ &= -1 + i \cdot 0 \\ &= -1 \end{aligned}$$

$$\therefore e^{i\pi} + 1 = 0 \text{ (Euler's identity)}$$

Example:

Example: $\begin{bmatrix} 0 & 63 \\ 127 & 255 \end{bmatrix}$

$$f(u,v) = \sum_{n=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cdot e^{-i2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

$$f(0,0) = \sum_{x=0}^1 \sum_{y=0}^1 f(x,y) \cdot e^{-i2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

$$= \left(0 \times e^{-i2\pi \left(\frac{0 \times 0}{2} + \frac{0 \times 0}{2} \right)} + 63 \times e^{-i2\pi \left(\frac{0 \times 0}{2} + \frac{0 \times 1}{2} \right)} + 127 \times e^{-i2\pi \left(\frac{0 \times 1}{2} + \frac{0 \times 0}{2} \right)} + 255 \times e^{-i2\pi \left(\frac{0 \times 1}{2} + \frac{0 \times 1}{2} \right)} \right)$$

$$= (0 \times 1 + 63 \times 1 + 127 \times 1 + 255 \times 1)$$

$$= 445$$

$$f(0,1) = \sum_{x=0}^1 \sum_{y=0}^1 f(x,y) \cdot e^{-i2\pi \left(\frac{0 \times 0}{2} + \frac{1 \times 0}{2} \right)} + 63 \times e^{-i2\pi \left(\frac{0 \times 0}{2} + \frac{1 \times 1}{2} \right)} + 127 \times e^{-i2\pi \left(\frac{0 \times 1}{2} + \frac{1 \times 0}{2} \right)} + 255 \times e^{-i2\pi \left(\frac{0 \times 1}{2} + \frac{1 \times 1}{2} \right)}$$

$$= (0 \times 1 + 63 \times (-1) + 127 + 255 \times (-1))$$

$$= -191$$

$$f(1,0) = \left(0 \times e^{-i2\pi \left(\frac{1 \times 0}{2} + \frac{0 \times 0}{2}\right)} + 63 \times e^{-i2\pi \left(\frac{1 \times 0}{2} + \frac{0 \times 1}{2}\right)} + 127 \times e^{-i2\pi \left(\frac{1 \times 1}{2} + \frac{0 \times 0}{2}\right)} + 255\right)$$

$$= (0 \times 1 + 63 \times 1 + 127 \times (-1) + 255 \times (-1))$$

$$= -319$$

$$f(1,1) = \left(0 \times e^{-i2\pi \left(\frac{1 \times 0}{2} + \frac{1 \times 0}{2}\right)} + 63 \times e^{-i2\pi \left(\frac{1 \times 0}{2} + \frac{1 \times 1}{2}\right)} + 127 \times e^{-i2\pi \left(\frac{1 \times 1}{2} + \frac{0 \times 1}{2}\right)} + 255 \times e^{-i2\pi \left(\frac{1 \times 1}{2} + \frac{1 \times 0}{2}\right)}\right)$$

$$= (0 - 63 - 127 + 255)$$

$$= 65$$

$$F(u,v) = \begin{bmatrix} 445 & -191 \\ -319 & 65 \end{bmatrix}$$

Inverse Discrete Fourier Transformation

Formula:

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) * e^{i2\pi(xu/M + yv/N)}$$

Example:

Example: $\begin{bmatrix} 445 & -191 \\ -319 & 65 \end{bmatrix}$

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} f(u,v) e^{i2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

$$f(0,0) = \frac{1}{2 \times 2} \sum_{u=0}^{1} \sum_{v=0}^{1} f(u,v) \cdot e^{i2\pi \left(\frac{ux}{2} + \frac{vy}{2} \right)}$$

$$= \frac{1}{4} (445 e^{i2\pi \left(\frac{0 \times 0}{2} + \frac{0 \times 0}{2} \right)} + (-191) e^{i2\pi \left(\frac{0 \times 0}{2} + \frac{0 \times 1}{2} \right)} + (-319) e^{i2\pi \left(\frac{0 \times 1}{2} + \frac{0 \times 0}{2} \right)} + 65)$$

$$= \frac{1}{4} (445 \times 1 + (-191 \times 1) + (-319 \times 1) + 65 \times 1)$$

$$= \frac{1}{4} \times 0 = 0$$

$$f(0,1) = \frac{1}{2 \times 2} (445 e^{i2\pi \left(\frac{0 \times 0}{2} + \frac{0 \times 1}{2} \right)} + (-191) e^{i2\pi \left(\frac{0 \times 0}{2} + \frac{1 \times 1}{2} \right)} + (-319) e^{i2\pi \left(\frac{0 \times 1}{2} + \frac{0 \times 0}{2} \right)} + 65)$$

$$= \frac{1}{4} (445 \times 1 + 191 - 319 - 65)$$

$$= \frac{1}{4} (252) = 63$$

$$f(1,0) = \frac{1}{4} \left(445 \times e^{i2\pi \left(\frac{0 \times 0}{2} + \frac{0 \times 0}{2} \right)} + (-191) \times e^{i2\pi \left(\frac{1 \times 0}{2} + \frac{0 \times 1}{2} \right)} + (-319) \times e^{i2\pi \left(\frac{1 \times 1}{2} + \frac{0 \times 0}{2} \right)} + 65 \times e^{i2\pi \left(\frac{1 \times 1}{2} + \frac{0 \times 1}{2} \right)} \right)$$

$$= \frac{1}{4} (445 - 191 + 319 - 65)$$

$$= \frac{1}{4} \times 508 = 127 //$$

$$f(1,1) = \frac{1}{4} \left(445 \times e^{i2\pi \left(\frac{1 \times 0}{2} + \frac{1 \times 0}{2} \right)} + (-191) \times e^{i2\pi \left(\frac{1 \times 0}{2} + \frac{1 \times 1}{2} \right)} + (-319) \times e^{i2\pi \left(\frac{1 \times 1}{2} + \frac{1 \times 0}{2} \right)} + 65 \times e^{i2\pi \left(\frac{1 \times 1}{2} + \frac{1 \times 1}{2} \right)} \right)$$

$$= \frac{1}{4} (445 + 191 + 319 + 65)$$

$$= \frac{1}{4} \times 1020 = 255 //$$

$$F(n,y) = \begin{bmatrix} 0 & 63 \\ 127 & 255 \end{bmatrix} //$$

Quantum Fourier Transformation(QFT)

- ◆ Definition:
 - The Quantum Fourier Transform (QFT) is the quantum version of the Discrete Fourier Transform (DFT).
 - It transforms a quantum state into a new quantum state in the frequency domain.

- Formula:

$$\text{QFT } |x\rangle^n = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega^{xy} |y\rangle^n$$

(or)

$$\text{QFT } |x\rangle^n = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{\frac{2\pi i xy}{N}} |y\rangle^n$$

Example: To apply QFT on classical matrix, we must encode the give data into quantum state or superposition state by amplitude encoding.

Amplitude Encoding: The process of flattening & normalizing the give matrix is known as amplitude encoding.

Quantum Fourier Transformation (QFT)

To apply the QFT on the classical matrix, we must :

- (1) Encode the data into the quantum state.
- (2) Apply QFT on that quantum state.
- (3) Compute and show the output Amplitudes.

Step-1: Input Matrix

$$X = \begin{bmatrix} 0 & 63 \\ 127 & 255 \end{bmatrix}$$

Treat it as 1D Array of size $N=4$

Step-2: Normalize and Encode as Quantum State

$$\begin{aligned} \text{Let } \vec{x} &= \begin{bmatrix} 0 \\ 63 \\ 127 \\ 255 \end{bmatrix} \Rightarrow \text{Normalize factor} = \sqrt{0^2 + 63^2 + 127^2 + 255^2} \\ &= \sqrt{0 + 3969 + 16129 + 65025} \\ &= \sqrt{85123} \\ &\approx 291.82 \end{aligned}$$

So the Normalized quantum state is:

$$|\psi\rangle = \frac{1}{291.82} \begin{bmatrix} 0 \\ 63 \\ 127 \\ 255 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.216 \\ 0.435 \\ 0.874 \end{bmatrix}$$

Step-3: QFT formula

where $N=2^n$ (here $N=4$)

$$x_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \cdot e^{2\pi i \cdot j \cdot k / N}$$

$$\text{So } \frac{1}{\sqrt{N}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$\begin{aligned} \rightarrow x_0 &= \frac{1}{2} \sum_{j=0}^3 x_j \cdot e^{\frac{2\pi i \cdot j \cdot 0}{4}} = \frac{1}{2} [x_0 + x_1 + x_2 + x_3] = \frac{1}{2} [0 + 0.216 + 0.435 + 0.874] \\ &= \frac{1}{2} [1.525] = 0.7625 \end{aligned}$$

$$= \frac{1}{2} \left[1.525 \right] = 0.7625$$

$$\rightarrow x_1 = \frac{1}{2} \sum_{j=0}^3 x_j \cdot e^{\frac{2\pi i \cdot j \cdot 1}{4}} = \frac{1}{2} [x_0 e^0 + x_1 e^{2\pi i \cdot \frac{1}{4}} + x_2 e^{2\pi i \cdot \frac{2}{4}} + x_3 e^{2\pi i \cdot \frac{3}{4}}]$$

$$e^{2\pi i \cdot \frac{1}{4}} = e^{i\pi/2} = i$$

where,

$$e^{2\pi i \cdot \frac{2}{4}} = e^{i\pi} = -1$$

$$e^{2\pi i \cdot \frac{3}{4}} = e^{3i\pi/2} = -i$$

$$= \frac{1}{2} [0 + 0.216i + (-0.435) + (-0.874i)]$$

$$= \frac{1}{2} [-0.435 - 0.658i] = -0.2175 - 0.329i$$

$$\rightarrow x_2 = \frac{1}{2} \sum_{j=0}^3 x_j \cdot e^{\frac{2\pi i \cdot j \cdot 2}{4}}$$

$$= \frac{1}{2} [x_0 e^0 + x_1 e^{i\pi} + x_2 e^{2i\pi} + x_3 e^{3i\pi}]$$

where $e^{\frac{2\pi i \cdot 1}{2}} = -1$

$$e^{\frac{2\pi i \cdot 2}{2}} = 1$$

$$e^{\frac{2\pi i \cdot 3}{2}} = -1$$

$$= \frac{1}{2} (0 - 0.216 + 0.435 - 0.874) = \frac{1}{2} (-0.655)$$

$$= -0.3275$$

$$\rightarrow x_3 = \frac{1}{2} \sum_{j=0}^3 x_j \cdot e^{\frac{2\pi i \cdot j \cdot 3}{4}}$$

$$e^{\frac{2\pi i \cdot 3}{4}} = -i$$

$$e^{\frac{2\pi i \cdot 6}{4}} = -1$$

$$e^{\frac{2\pi i \cdot 9}{4}} = i$$

$$= \frac{1}{2} [0 + 0.216(-i) + 0.435(-1) + 0.874(i)]$$

$$= \frac{1}{2} [-0.435 + (0.874 - 0.216)i] = \frac{1}{2} [-0.435 + 0.658i]$$

$$= -0.2175 + 0.329i$$

Step-4:

Resultant state vector

$z =$

$$\begin{bmatrix} 0.7625 \\ -0.2175 - 0.329i \\ -0.3275 \\ -0.2175 + 0.329i \end{bmatrix}$$

Inverse Quantum Fourier Transformation

The inverse QFT formula is:

$$f_n = \frac{1}{\sqrt{N}} \sum_{p=0}^{N-1} F_p \cdot e^{-2\pi i \cdot \frac{np}{N}} \quad \text{where } N = 4$$

Example:

Inverse QFT:

$$\text{Result of QFT} = \begin{bmatrix} 0.7625 & \\ -0.2175 - 0.329i & \\ -0.3275 & \\ -0.2175 + 0.329i & \end{bmatrix} = \begin{bmatrix} F_0 \\ F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

Step-1: Inverse QFT formula is:

$$f_n = \frac{1}{\sqrt{N}} \sum_{p=0}^{N-1} F_p \cdot e^{-2\pi i \cdot np / N} \quad \text{where } N=4$$

Step-2: Compute f_n (compute Normalized values)

$$\begin{aligned} \rightarrow f_0 &= \frac{1}{2} [F_0 + F_1 + F_2 + F_3] \\ &= \frac{1}{2} [0.7625 - 0.2175 - 0.329i - 0.3275 - 0.2175 + 0.329i] \\ &= \frac{1}{2} (0) = \boxed{0} \end{aligned}$$

$$\begin{aligned} \rightarrow f_1 &= \frac{1}{2} [F_0 + F_1 \cdot e^{-2\pi i \cdot 1 \cdot \frac{1}{4}} + F_2 \cdot e^{-2\pi i \cdot 1 \cdot \frac{2}{4}} + F_3 \cdot e^{-2\pi i \cdot 1 \cdot \frac{3}{4}}] \\ &= \frac{1}{2} [0.7625 - 0.2175 - 0.329i - 0.3275 - 0.2175 + 0.329i] \\ &= \frac{1}{2} (0) = \boxed{0} \end{aligned}$$

$$e^{-2\pi i \cdot 1 \cdot \frac{1}{4}} = e^{-\frac{j\pi}{2}} = -i$$

$$e^{-2\pi i \cdot 1 \cdot \frac{2}{4}} = -1$$

$$e^{-2\pi i \cdot 1 \cdot \frac{3}{4}} = i$$

$$f_1 = \frac{1}{2} \left[\underbrace{0.7625 + (-0.2175 - 0.329i)(-i)}_{\Rightarrow 0.329 + 0.2175i} + \underbrace{(-0.3275)(-1) + (-0.2175 + 0.329i)(i)}_{\Rightarrow -0.329 - 0.2175i} \right]$$

$$= \frac{1}{2} [0.7625 - 0.329 + 0.2175i + 0.3275 - 0.329 - 0.2175i]$$

$$= \frac{1}{2} (0.432) = \boxed{0.216}$$

$$\rightarrow f_2 = \frac{1}{2} [f_0 + f_1(-1) + f_2(1) + f_3(-1)]$$

$$= \frac{1}{2} [0.7625 + 0.2175 + 0.329i - 0.3275 + 0.2175 - 0.329i]$$

$$= \frac{1}{2} [0.7625 + 0.2175 - 0.3275 + 0.2175]$$

$$= \frac{1}{2} [0.87] = \boxed{0.435}$$

$$\rightarrow f_3 = \frac{1}{2} [f_0 + f_1(i) + f_2(-1) + f_3(-i)]$$

By Calculating, we got

$$= \boxed{0.875}$$

Step-3: Inverse GFI Result:

$$f_n = \begin{bmatrix} 0 \\ 0.216 \\ 0.435 \\ 0.874 \end{bmatrix}$$

Step-4: finding the Norm:

choose the largest value from the above result
i.e., $\boxed{0.874}$

→ As the given example matrix is '8 bits per pixel image and largest possible value for 8 bits is '255'

→ Now divide 255 by 0.874

→ By calculating, we get Norm = $\frac{255}{0.874} \approx 291.76$

→ By calculating, we get Norm = $\frac{255}{0.874}$

Step-5: Multiplying with Norm:

$$\Rightarrow f_n \times \text{Norm}$$

$$f_0 \Rightarrow 0 \times 291.76 = 0$$

$$f_1 \Rightarrow 0.81 \times 291.76 \approx 236$$

$$f_2 \Rightarrow 0.435 \times 291.76 \approx 127$$

$$f_3 \Rightarrow 0.874 \times 291.76 \approx 255$$

$$\text{Resultant Matrix} = \begin{bmatrix} 0 & 236 \\ 127 & 255 \end{bmatrix}$$

**Thank
You...!!**